

MLCC 2026

Lab 2 – RLS with linear models

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Regularized Least Squares

$$\min_{w \in \mathbb{R}^D} \frac{1}{n} \sum_{i=1}^n (y_i - w^T x_i)^2 + \lambda w^T w, \quad \lambda \geq 0$$

$$f(x) = x^T w, \quad \forall x \in \mathbb{R}^D$$

$$(X_n^T X_n + \lambda n I) w = X_n^T Y_n$$

Regularized Least Squares

$$\underbrace{(X_n^T X_n + \lambda n I)}_A w = \underbrace{X_n^T Y_n}_b$$

$$Aw = b$$

Regularized Least Squares

```
regularizedLSTrain(Xtr, Ytr, lam)
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$$(X_n^T X_n + \lambda n I) w = X_n^T Y_n$$

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return w
```

Regularized Least Squares

regularizedLSTest(w, Xte)

$$f(x) = x^T w$$

return Ypred

Regularized Least Squares

Goals:

- Implement and use RLS for regression
- Play with noise and regularization parameter
- K-Fold Cross Validation